

math-devcontainer

A devcontainer to build a grad school math course workspace. Opioninated for Python, Latex, and Markdown.

How to use the devcontainer

I generally use this workspace in one of two ways.

- Local development on VS Code using either of my machines (Macbook or Linux box)
- Codespaces on Github (which allows me to work on any machine I can login to Github, including my iPad)

How I structure coursework

I assume that the course is structured in weekly modules and that course work involves some problem set from a textbook and some software assignment.

First, I create a directory called `module- $\{x\}$` where x is that week's module number.

Then, I create a markdown like `assignment-example.md`. I generally use inline latex in markdown for all the problem sets, see the example assignment for more details.

The devcontainer includes a full LaTeX toolchain (TeX Live) plus ChkTeX for linting LaTeX source. The previous automatic markdown→PDF on save via pandoc has been removed to keep the image lean and simplify dependencies. You can still export Markdown to PDF manually (e.g. using VS Code's Markdown PDF extensions or an external pandoc install on your host) if desired.

To lint a LaTeX file you can run:

```
chktex assignment-example.tex
```

For any part of the assignment that uses Python, I change directory to the module and then use I use uv which is already installed on the devcontainer to set up Python for each module.

1. `cd module-example`
2. `uv init`
3. `uv add numpy matplotlib` or whatever dependencies you need.
4. `uv run software_assignment.py`
5. [Optional for linting] `uv add ruff`
6. [Optional for linting] `uv run -m ruff format`

I then copy the Python code into the assignment and link any images created from a library like `matplotlib`. See the example assignment for more details.